Software Requirements Specification (SRS)

ASPIRE: After-School Program Integrated Resource & Education System — Parent Module

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1. Introduction

1.1 Purpose

The purpose of this document is to describe the software requirements for the Parent Module of ASPIRE, a multi-center after school program platform. It explains how parents can follow their child's progress, attendance, assignments, fee invoices and payments, and receive notifications about their child's activities.

This document will help developers, testers and project members understand the expected system behavior and build the module according to the requirements.

1.2 Intended Audience

This document is intended for:

- **Developers:** to understand and implement the required features.
- **Testers:** to verify that each feature works correctly using the confirmation criteria.
- **Project manager and team members:** to monitor progress and keep the Parent Module consistent with the Teacher, Student, Center Admin and Super Admin modules.
- **UI/UX designers:** to design simple, mobile-friendly screens for parents.
- **Center Admin and Super Admin:** to understand what parents can see and what data they must keep accurate.

1.3 Intended Use

This SRS will be used as a reference during system design, development, testing and maintenance. Developers should follow the requirements when implementing the Parent Module, and testers should use the confirmation criteria to check that each feature works as expected. Any change to a requirement must be agreed by the team and recorded in a new version.

1.4 Product Scope

The Parent Module is a part of ASPIRE. It gives parents a single place to stay informed about their child's learning and fees. The module will allow parents to:

- Log in and see a dashboard for each linked child.
- View marks, grade breakdown, level progress and teacher feedback.
- View attendance records and attendance percentage.

- Track assignments and their status.
- View fee invoices and download them.
- View payment history and download receipts.
- Receive and manage notifications about their child's activities.

The parent has **read-only** access. Parents do not submit work, mark attendance, edit results or record payments.

The main goal is to improve transparency between the center and the family, help parents act early when a child is falling behind or a fee is due, and reduce the manual work of center staff answering parent queries.

Out of scope for this module: online payment, direct messaging between parents and teachers, and editing any academic or financial record.

1.5 Risk Definitions

The following risks may affect the module if the requirements are not implemented correctly.

ID	Risk	Possible impact	Related requirements
R1	A parent sees the data of a child who is not linked to them (another family).	Privacy breach and loss of trust.	FR-01, NFR-02, NFR-03
R2	A parent is linked to the wrong child.	A family sees another child's data.	FR-01, NFR-02
R3	A parent sees draft assignments or unpublished grades and feedback.	Wrong or confusing information.	FR-02, FR-04, NFR-02
R4	Important notifications (absence, overdue assignment, new invoice) are missed or delayed.	Parents miss deadlines, payments or warning signs.	FR-07, NFR-05
R5	Invoice, payment or balance information shown to the parent is wrong.	Fee disputes and financial loss.	FR-05, FR-06, NFR-04
R6	Attendance or marks shown are outdated or incorrect.	Wrong decisions about the child.	FR-02, FR-03, NFR-04
R7	Account takeover through a weak or guessed password.	Exposure of a child's personal and financial data.	FR-01, NFR-02

2. Overall Description

The Parent Module is part of a new product. ASPIRE is being built because centers currently have no single place to manage curriculum, attendance, grading and fees, and parents have little visibility into their child's learning. The Parent Module is built for the guardians who enroll the child and pay the fees.

2.1 User Classes and Characteristics

Parent (primary user of this module and the buyer). The guardian of one or more students. Parents have a wide range of technical skill and mostly use a phone browser. They use the system weekly or when they receive a notification. A parent may have more than one linked child.

Student. The child whose data the parent views. The student's actions (submissions, attendance, results) produce the data that the parent sees.

Teacher. Takes attendance, publishes assignments, grades and feedback, and notifies parents. The teacher's data and actions feed the Parent Module.

Center Admin. Links a parent to a student, generates fee invoices, records payments and manages the center's records. The invoice and payment data shown to parents comes from the Center Admin.

Super Admin. Manages centers and all user accounts, and can link or unlink a parent and a student.

2.2 User Needs

The parent needs to:

- Know at a glance whether the child attends class and completes homework.
- Understand the child's marks, level progress and the teacher's feedback.
- See clearly which fees are due, how much, and by when.
- Have proof of payment (receipts) and a record of what has been paid.
- Be notified about important events without having to ask the center.
- Use the system easily on a phone, even with limited technical skill.
- Be sure that only they can see their child's private information.

2.3 Operating Environment

- **Platform:** web-based application.
- **Devices:** smartphones (primary), tablets, laptops and desktop computers.
- **Browsers:** modern versions of Chrome, Firefox, Edge and Safari.
- **Internet:** an active internet connection is required.
- **Backend:** a server-side application that handles authentication, permissions, academic and financial records and business rules.
- **Database:** stores users, parent-child links, attendance, assignments, grades, feedback, invoices, payments and notifications.
- **Notification services:** an email service (and in-app notifications) must be configured for notifications to be delivered.

- **Time:** server time is UTC; the interface shows Bangladesh time (UTC+6).

The exact technology stack will be decided in the System Design Specification.

2.4 Constraints

1. A parent can access only the data of children linked to their account.
2. The parent has read-only access; the module cannot be used to edit records.
3. Grades, feedback and assignments are visible only after the teacher has published them. Drafts are never visible.
4. All authorization must be enforced on the backend, not only by hiding buttons in the interface.
5. Parent-child linking is done only by the Center Admin or Super Admin.
6. Online payment is not supported; payments are recorded by the Center Admin.
7. Notification delivery depends on the configured email and notification services.
8. Personal data of children must be visible only to the linked parent, the relevant teachers, the center staff and the Super Admin.

2.5 Assumptions

- Every parent has a registered account with a valid email address, created or invited by the center.
- The Center Admin or Super Admin links each parent to their child correctly, and a parent may have more than one child.
- Teachers publish grades, feedback and assignments in a timely way, and take attendance for every class session.
- The Center Admin generates invoices and records payments promptly, so the parent sees accurate balances.
- The Teacher, Student and Center Admin modules generate the events that trigger parent notifications (absence, overdue assignment, published grade or feedback, new invoice, payment recorded, announcement) in a consistent way. Teachers can also notify parents directly.
- Parents have a smartphone or computer with an internet connection.
- If any of these assumptions turns out to be false, the affected requirements must be revised.

3. Requirements

3.1 Functional Requirements

FR-01: Login and Dashboard (US-P1)

User Story

As a parent, I want to log in and see a dashboard for my child (or children), so that I can quickly see how they are doing.

Confirmation / Acceptance Criteria

- The parent can log in using a valid email and password (the common login rules apply).
- The system rejects invalid credentials with the message "Invalid email or password" and does not say which one is wrong.
- If more than one child is linked, the parent can switch between children from the dashboard.
- The dashboard shows each child's attendance rate, pending assignments, latest grade, outstanding fees and recent notifications.
- A parent with no linked child sees a message to contact the center.
- The parent can see only the children linked to their account.
- The parent can log out securely from any page.

FR-02: View Child Progress (US-P2)

User Story

As a parent, I want to see my child's marks, level progress and teacher feedback, so that I can understand how my child is performing in each subject.

Confirmation / Acceptance Criteria

- The parent can view results per subject and per assessment only after the teacher has published them.
- Unpublished grades show "Unpublished".
- The parent can see the grade breakdown by category (for example Homework, Quizzes, Exams) and the overall progress bar.
- The parent can read the teacher's feedback on assignments and results.
- The parent can see a summary of overall performance, such as average marks and completed versus missed tasks.
- Feedback of other students is never visible to the parent.

FR-03: View Child Attendance (US-P3)

User Story

As a parent, I want to see my child's attendance record, so that I know if my child is missing classes.

Confirmation / Acceptance Criteria

- The parent can view attendance by date, course and status (Present, Absent, Late, Excused).
- The parent can see the attendance percentage and filter by date range.
- Low attendance is highlighted visibly.
- Attendance is read-only for the parent.
- If no attendance record exists for the selected filter, the system shows a clear empty message.

FR-04: Track Assignment Status (US-P4)

User Story

As a parent, I want to see my child's assignments and their status, so that I can make sure homework is completed on time.

Confirmation / Acceptance Criteria

- The parent sees each assignment's course, title, due date, marks and status (Pending, Submitted, Overdue).
- Overdue assignments are displayed in a separate, highlighted area.
- The parent receives a notification when an assignment becomes overdue.
- Draft assignments are never visible.
- The parent can view but cannot submit or edit anything for the child.
- The parent can filter assignments by status and course.

FR-05: View Fee Invoices (US-P5)

User Story

As a parent, I want to see my child's fee invoices, so that I know how much to pay and by when.

Confirmation / Acceptance Criteria

- The parent can see each invoice with invoice ID, course, billing period, total fee, discount, amount payable and due date.
- The parent can view the invoice status (Paid, Partially Paid, Due, Overdue).
- The parent can download an invoice as a PDF.
- The parent can see only the invoices of their own linked children.
- Overdue invoices are clearly highlighted.

FR-06: View Payment Records (US-P6)

User Story

As a parent, I want to see my payment history and receipts, so that I can verify what I have paid and what is still due.

Confirmation / Acceptance Criteria

- The parent can see payment date, amount, method, transaction ID and remaining balance for each payment.
- Partial payments are shown with the updated outstanding balance.
- The parent can download a receipt for any recorded payment.
- The parent cannot edit or delete payment records.
- The parent can see only the payments of their own linked children.

FR-07: Receive Notifications (US-P7)

User Story

As a parent, I want to get notifications about my child's activities, so that I do not miss important updates.

Confirmation / Acceptance Criteria

- The parent is notified of absence, overdue assignment, newly published grade or feedback, new invoice, upcoming due date, payment recorded, and teacher or center announcements.
- The parent can see a notification list, mark items as read and filter by type.
- The parent can choose which notification types to receive (in-app or email).
- Notifications are sent only to the parents of the relevant child.
- A notification failure does not affect the original record (for example the grade or invoice).

3.2 Non-Functional Requirements

NFR-01: Performance

- The dashboard and child pages should load within 3 seconds under normal operating conditions.
- Lists such as assignments, invoices and payments should be paginated so that large histories load quickly.
- The system should support many parents using it at the same time without significant slowdown.

NFR-02: Security

- Parents must authenticate before accessing any protected feature.
- Passwords must be stored in hashed form (bcrypt) and never saved in plain text.
- After 5 wrong password attempts within 15 minutes, the account is locked for a period of time.
- The backend must check on every request that the parent is linked to the requested child. Hiding buttons in the frontend is not enough.
- Role-based access control must be applied to all parent features.
- Unpublished grades, feedback and draft assignments must never be returned to the parent by the backend.
- All data must be transmitted over HTTPS.

NFR-03: Privacy and Confidentiality

- A child's personal and academic data is visible only to the linked parent, the relevant teachers, the center staff and the Super Admin.
- Notifications must not expose a child's information to other families.
- Invoices, receipts and downloaded files must contain only the linked child's information.

NFR-04: Data Accuracy and Integrity

- Attendance, grades, invoices, payments and balances shown to the parent must match the actual recorded data.
- Payment records are read-only for parents and are never permanently deleted.
- Parent-child links must be changed only by authorized users, and each change must be logged.

NFR-05: Reliability and Notifications

- Notifications should be delivered within a few minutes of the event.
- If a request fails, the system should show an understandable error message.
- Delivery status of notifications should be tracked where supported.

NFR-06: Usability

- The interface should be simple, consistent and easy to understand for parents with limited technical skill.
- It should follow the Nielsen usability heuristics: visible system status, simple language and helpful error messages.
- Statuses such as Overdue, Low attendance and Unpublished should be visibly distinguished.
- Empty states (for example no overdue assignments) should show a clear message.

NFR-07: Compatibility and Responsiveness

- The application should work on supported modern web browsers.
- The layout should be mobile-first, since most parents will use a phone, and adapt to tablet and desktop screens.

NFR-08: Auditability

- The system should log parent access to sensitive records such as invoices, receipts and results.
- Audit records should identify the user, the affected record and the date and time, and ordinary users must not be able to change or delete them.

NFR-09: Time and Date Accuracy

- All timestamps are stored in UTC on the server and displayed in Bangladesh time.
- Due dates and overdue status must be calculated consistently across assignments and invoices.

NFR-10: Maintainability

- The module should be built with a clear structure (for example MVC) so features can be updated without affecting other modules.
- Functional requirements should be testable, and errors should be logged without exposing sensitive information.